UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	INTERACTIVE MULTIMEDIA
COURSE CODE	:	CSC413
EXAMINATION	:	NOV 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC413 mock paper (NOV 2026). Answers show the expected marking points; all questions are deterministic and paper-derived; no lecture slides are used.

2. Marking points show the expected answer and the allocation of marks; unresolved slots would remain visible.

3. Part A = 15 x 1 mark; Part B = 15 x 1 mark; Part C = 30 x 1 mark; Part D = 4 x 10 marks; total = 100.

4. Questions: 64. Slots: 72. Placeholders: 0. No LLM call. Questions and answers are for local study use only.

5. Structure derived from 15 unique CSC413 past papers; no past-paper text is copied.

FOR MARKERS' USE ONLY

This marking scheme consists of 6 printed pages

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PART A - ANSWER KEY

Q	Ans	Statement	Explanation
A1	TRUE	Authoring tools help developers organise and edit multimedia elements.	Authoring tools assemble media.
A2	TRUE	A URL identifies the address of a web resource.	A URL is a resource locator.
A3	TRUE	HTML is used to structure content on web pages.	HTML marks up web content.
A4	TRUE	RGB is an additive colour model.	RGB adds red, green and blue light.
A5	TRUE	Sampling converts a continuous sound wave into digital values.	Sampling measures the wave at intervals.
A6	FALSE	Interlacing displays every line of a video frame at the same time.	Interlacing draws alternate lines.
A7	TRUE	A scanner is an input device.	A scanner captures documents into the computer.
A8	TRUE	Streaming allows media to begin playing before the whole file is downloaded.	Streaming delivers media continuously.
A9	TRUE	A prototype is an early model of a product.	Prototypes test ideas before production.
A10	TRUE	Tweening creates intermediate frames between keyframes.	Tweening fills the gaps.
A11	TRUE	Interactivity gives users control over content and flow.	Interactivity is a defining feature of many multimedia products.
A12	FALSE	Higher resolution always means a smaller file.	Higher resolution usually increases file size.
A13	FALSE	A plotter is an input device.	A plotter produces output.
A14	TRUE	The amplitude of a sound wave determines its loudness.	Larger amplitude means louder sound.
A15	TRUE	A higher sampling rate generally produces a more accurate recording.	More samples capture more detail.

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PART B - ANSWER KEY

Q	Ans	Term	Definition
B1	C	Bitmap	image made of pixels
B2	G	Virtual reality	simulated environment
B3	H	Storyboard	visual sequence plan
B4	O	Sampling	converting sound into digital values
B5	K	Serif	font with small strokes at character ends
B6	M	Card-based tool	authoring tool organised like pages or cards
B7	D	Target audience	intended users of a product
B8	E	Multimedia	integration of text, graphics, audio, video and animation
B9	I	Linear multimedia	presentation without user control of sequence
B10	B	MIDI	standard for musical notes and instructions
B11	F	Hypermedia	linked multimedia elements
B12	J	Interactivity	user control over content and flow
B13	L	Convergence	merging of computing, media and communications
B14	N	Scanner	input device that digitises documents
B15	A	Keyframe	important frame in an animation

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PART C - ANSWER KEY

Q	Ans	Correct answer
C1	C	Display web pages
C2	A	Number of pixels in an image
C3	D	It combines visuals and sound
C4	C	Links to other content
C5	B	Modelling and rendering
C6	B	Permission to use a work
C7	A	Along a predefined route
C8	D	Drawing each frame by hand
C9	D	Screen display
C10	C	Play while downloading
C11	B	Compiler
C12	A	Space between individual character pairs
C13	B	An early model of a product
C14	B	Messages with a beginning and end
C15	C	In-between frames
C16	A	A visual sequence plan
C17	B	Scanner
C18	A	Multimedia with hyperlinks
C19	B	Draws alternate lines of a frame
C20	B	Important frames in an animation
C21	D	CMYK
C22	D	Replace a colour background
C23	B	PNG
C24	C	Pixels
C25	A	Musical notes and instructions
C26	C	Temporary working memory
C27	C	A video compression family
C28	A	Frames per second
C29	A	Plotter
C30	D	A cinema presentation

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PART D - WORKED ANSWERS

QUESTION 1

Topic: authoring interactivity

A company wants an interactive product demonstration that runs on a kiosk and on the web.

a) State TWO features to consider when choosing authoring software.

Interactivity support and export formats are important features.

(2 marks)

b) Explain the interactivity level suitable for the demonstration.

A medium level with menus and buttons lets users explore without overwhelming them.

(4 marks)

c) Explain TWO authoring tool categories appropriate for this project.

Card-based tools suit structured screens; timeline-based tools suit animation and media sequencing.

(4 marks)

QUESTION 2

Topic: animation

You are required to animate a company logo. The logo depicts a planet orbiting the sun.

a) State TWO animation file formats suitable for this project.

GIF and MP4 are suitable; SWF is also acceptable.

(3 marks)

b) Describe the steps of cel animation and ONE advantage of using it.

Draw each frame, capture the frames in sequence, then play them back; it gives direct artistic control.

(3 marks)

c) Differentiate 2-D and 2.5-D animation with ONE example for each.

2-D uses flat drawings such as a cartoon; 2.5-D adds depth to flat artwork, such as parallax layers in a game.

(4 marks)

QUESTION 3

Topic: web hypermedia

A museum wants a website that presents its collection and lets visitors explore related exhibits.

a) State TWO differences between linear and non-linear navigation.

Linear follows a fixed sequence; non-linear lets users choose their own path.

(2 marks)

b) Describe TWO roles of hyperlinks in the website.

They connect related exhibits and let users move between pages.

(3 marks)

c) Explain how streaming and compression affect multimedia delivery on the web.

Compression reduces file size for faster transfer; streaming lets audio or video play while it downloads.

(5 marks)

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QUESTION 4

Topic: design project

A shopping complex requires a kiosk that gives customers location and retailer information.

a) State TWO important project team roles for completing the kiosk.

A project manager and an interface designer are important roles.

(3 marks)

b) Design a low-fidelity prototype for a pre-school courseware home page. The page must show topic menus, the course title and the school logo.

A simple sketch with a title, logo and large topic buttons is sufficient at low fidelity.

(3 marks)

c) Discuss TWO factors that lead to the wide use of mobile devices in games and shopping applications.

Portability and constant connectivity make mobile devices convenient for games and shopping.

(4 marks)

END OF MARKING SCHEME

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